

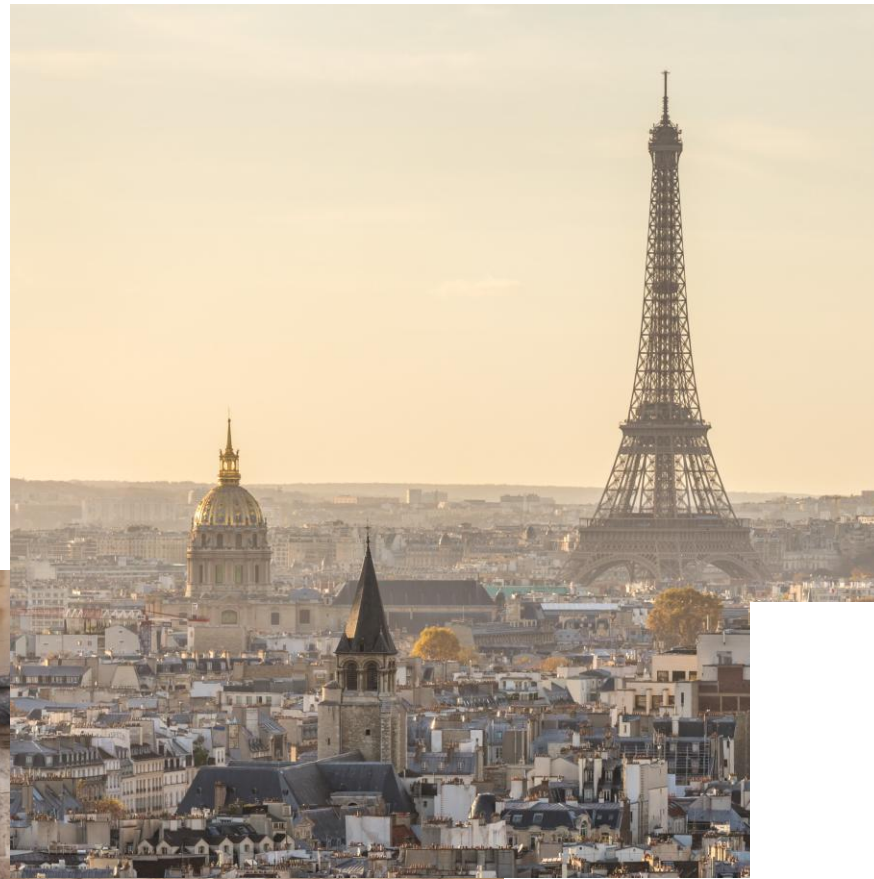
Microeconomics II



Topic 5: Firm supply under perfect competition

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About me



About you



Roadmap

1. Perfect competition
2. Profit maximisation
3. Firm supply curve
4. Profit and producer's surplus
5. Comparative statics

Motivation



Market environments

- 2 types of constraints on firm's choice of price and quantity:
 1. **Technological constraints** – given by the production function and leading to a cost function
 2. **Market constraints** – given by a demand curve
- Demand faced by a firm
 - If single firm: demand curve determined purely by consumer behaviour
 - If multiple firms: market constraints determined also by behaviour of other firms → the market environment
 - need to distinguish “demand faced by a firm” vs. “market demand”
- Types of market environments
 - If a small number of firms strategically interact, exact nature of this interaction matters
 - See the lecture on oligopoly and the Industrial Organisation course
 - If many small firms, analysis easier (and realistic in some contexts) → perfect (pure) competition

Perfect competition: assumptions

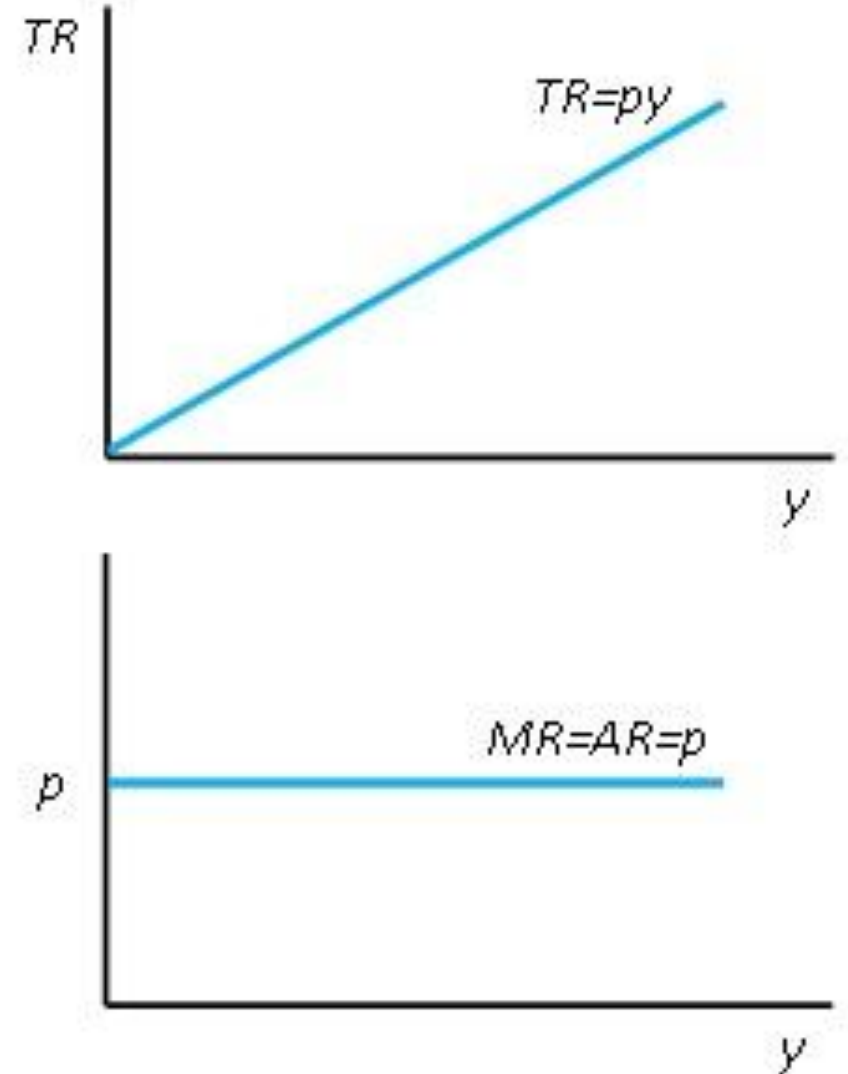
- Key assumption: firms are **price-takers**
 - firms are small relative to the market
 - market price independent of each firm's level of output
 - firms face horizontal demand curve
- Additional assumptions:
 - Free entry to and exit from the industry (in the long term)
 - Homogenous product (for heterogeneous products, see the lecture on oligopoly)
 - Supply of the input faced by a firm is also horizontal
 - Perfect information (assumed throughout Micro II, for imperfect information see the IO course)
 - (All firms identical – sometimes assumed but not required by perfect competition)

Examples of perfect competition



Revenues

- Firms maximise $\pi = TR - TC = py - \sum_{i=1}^N w_i x_i$
- **Revenues**
 - Total revenues
 - Generally: $TR(y) = p(y) * y$
 - Perfect competition: $TR(y) = p * y$
 - Marginal revenues
 - Rate by which revenue increases with a change in output
 - Generally: $MR(y) = \frac{dTR(y)}{dy} = \frac{dp(y)y}{dy} = p(y) + y \frac{dp(y)}{dy}$
 - Perfect competition: $MR(y) = p$
 - Average revenues
 - Revenues per unit of output
 - Generally: $AR(y) = \frac{TR(y)}{y} = \frac{p(y)y}{y} = p(y)$
 - Perfect competition: $AR(y) = p$



Profit maximisation: the problem

$$\max \pi(y, x_1, \dots, x_n) = TR(y) - \sum_{i=1}^N w_i x_i$$

such that $y \leq f(x_1, \dots, x_n)$; $y \geq 0$; $x_i \geq 0, i = 1, \dots, n$

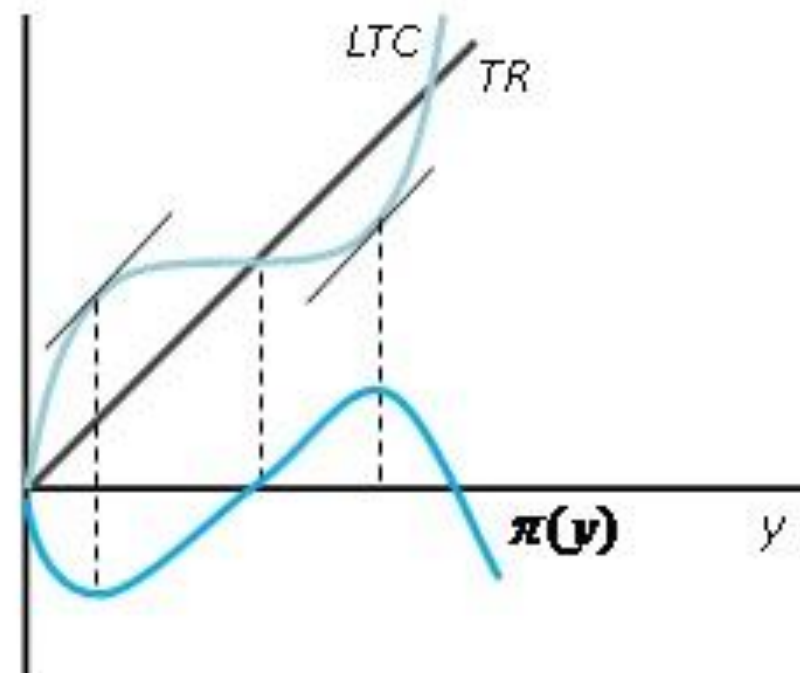
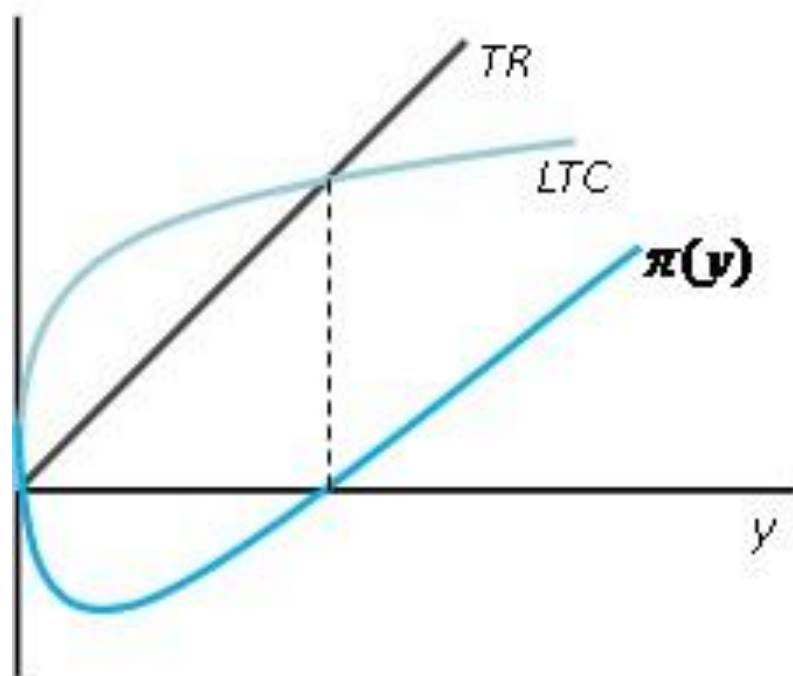
- Using the cost function derived earlier: $\max \pi(y) = TR(y) - c(w, y)$ s.t. $y \geq 0$
- In the case of perfect competition: $\max \pi(y) = py - c(w, y)$ s.t. $y \geq 0$

Profit maximisation: optimum conditions

- First order condition (FOC):
 - Additional units of output do not bring additional profit
 - Generally: $\frac{d\pi(y^*)}{dy} = \frac{dTR(y^*)}{dy} - \frac{dTC(y^*)}{dy} = 0$
 - Generally: $MR(y^*) = MC(y^*)$
 - Perfect competition: $p = MC(y^*)$
- Second order condition (SOC):
 - For the point to be maximum, marginal profit must be decreasing in output (profit a concave function)
 - Generally: $\frac{d^2\pi(y^*)}{dy^2} = \frac{d^2TR(y^*)}{dy^2} - \frac{d^2TC(y^*)}{dy^2} = \frac{dp}{dy} - \frac{dMC(y^*)}{dy} < 0$
 - Perfect competition: $\frac{dMC(y^*)}{dy} > 0 \rightarrow$ output on the upward-sloping portion of the MC curve

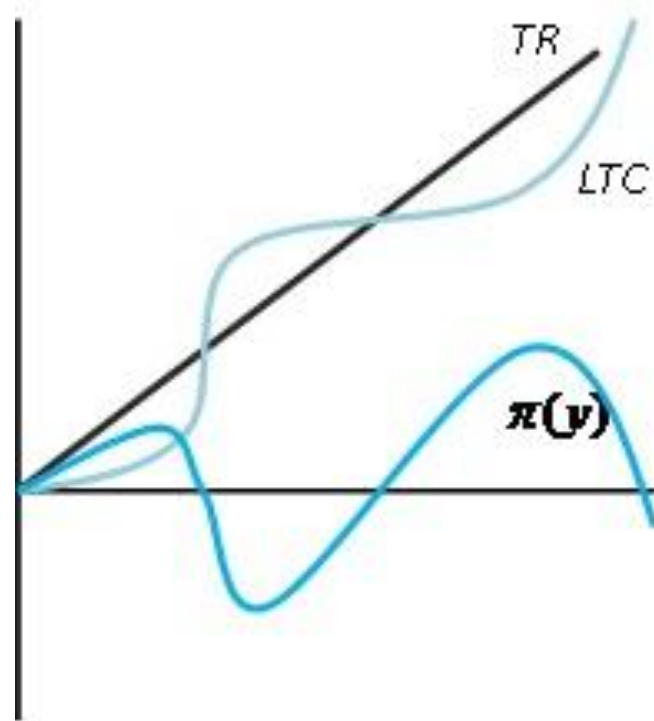
Profit maximisation: existence of maximum

In which of the cases below is there a profit maximum?



Profit maximisation: multiple maxima

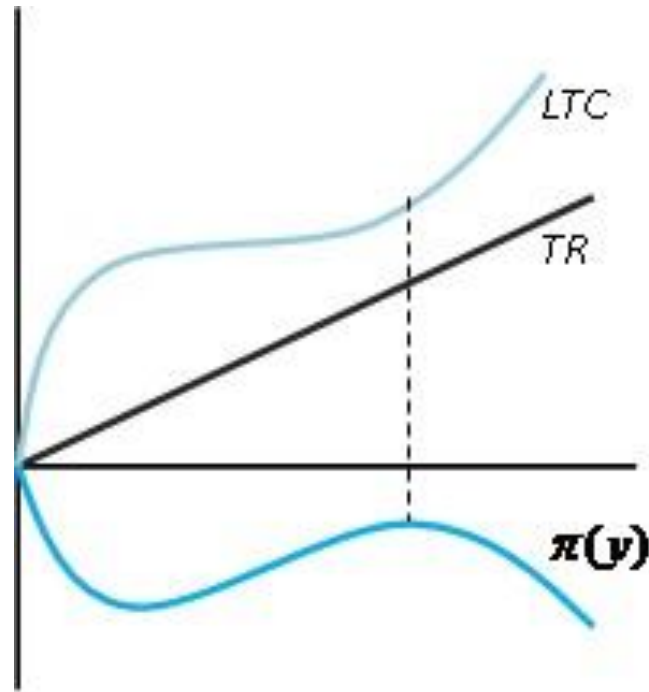
Can there be multiple local profit maxima?



- Shape of the profit function given by the shape of the cost function
- There may be more local maxima of the profit.
- Compare and choose the point with the highest profit.

Profit maximisation: multiple maxima

Can maximised profit be negative?



- Profits maximised in $y=0$, even though FOC and SOC not satisfied because $\frac{d\pi(0)}{dy} = p - \frac{MC(0)}{dy} < 0$
- The firm chooses not to produce and have zero profits

Profit maximisation: summary

- Profit in a local maximum
 - $a) \quad p = MC(y^*) \text{ and } \frac{dMC(y^*)}{dy} > 0 \text{ (FOC and SOC satisfied)}$
 - $b) \quad y = 0 \text{ (corner solution)}$
- There can be no maximum at all (not very realistic...)
- There can be multiple local maxima → maximum is the one with the highest profits

Profit maximisation: Exercise

Setup:

- $TC(y) = 2y^3 - 30y^2 + 150y$
- Perfect competition

Question:

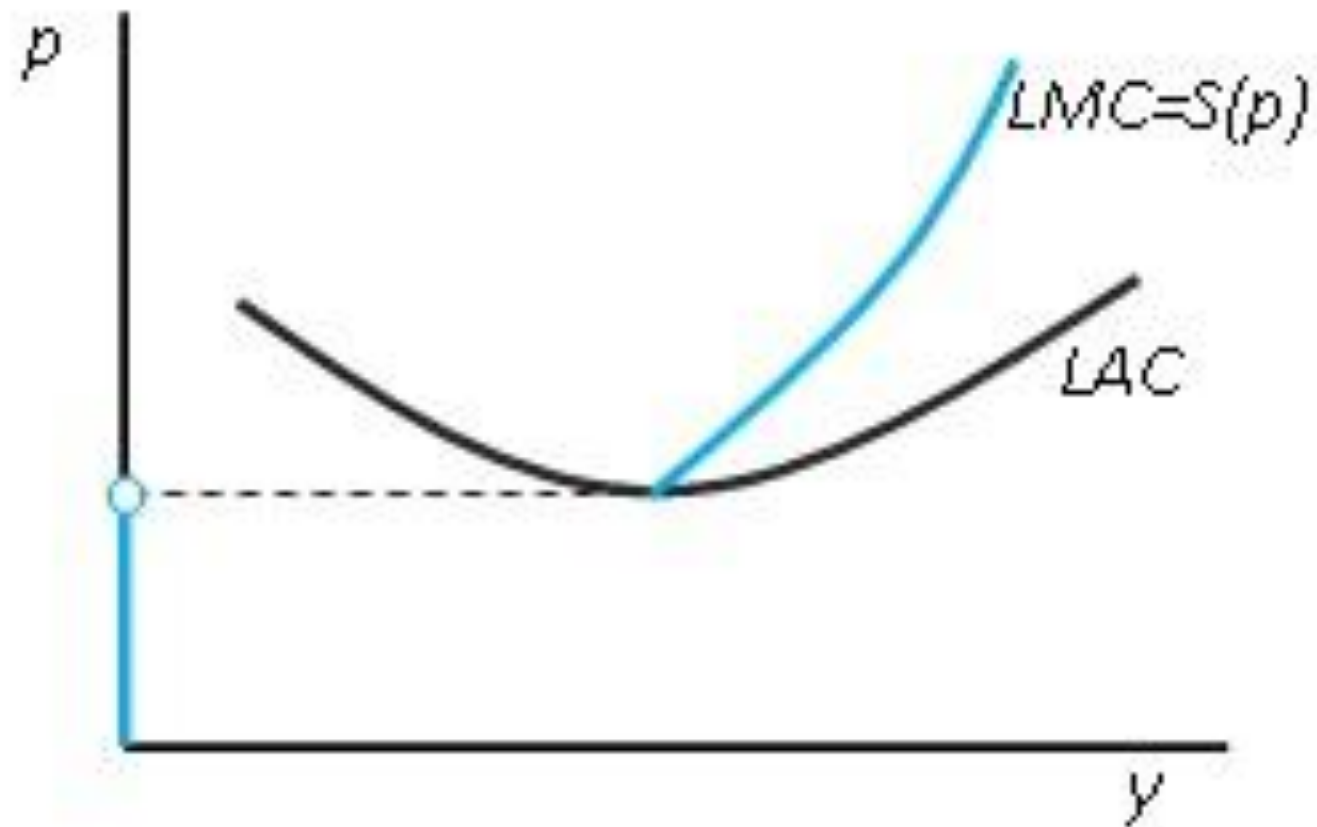
1. What output would be produced at price $p=6$?



Firm supply curve

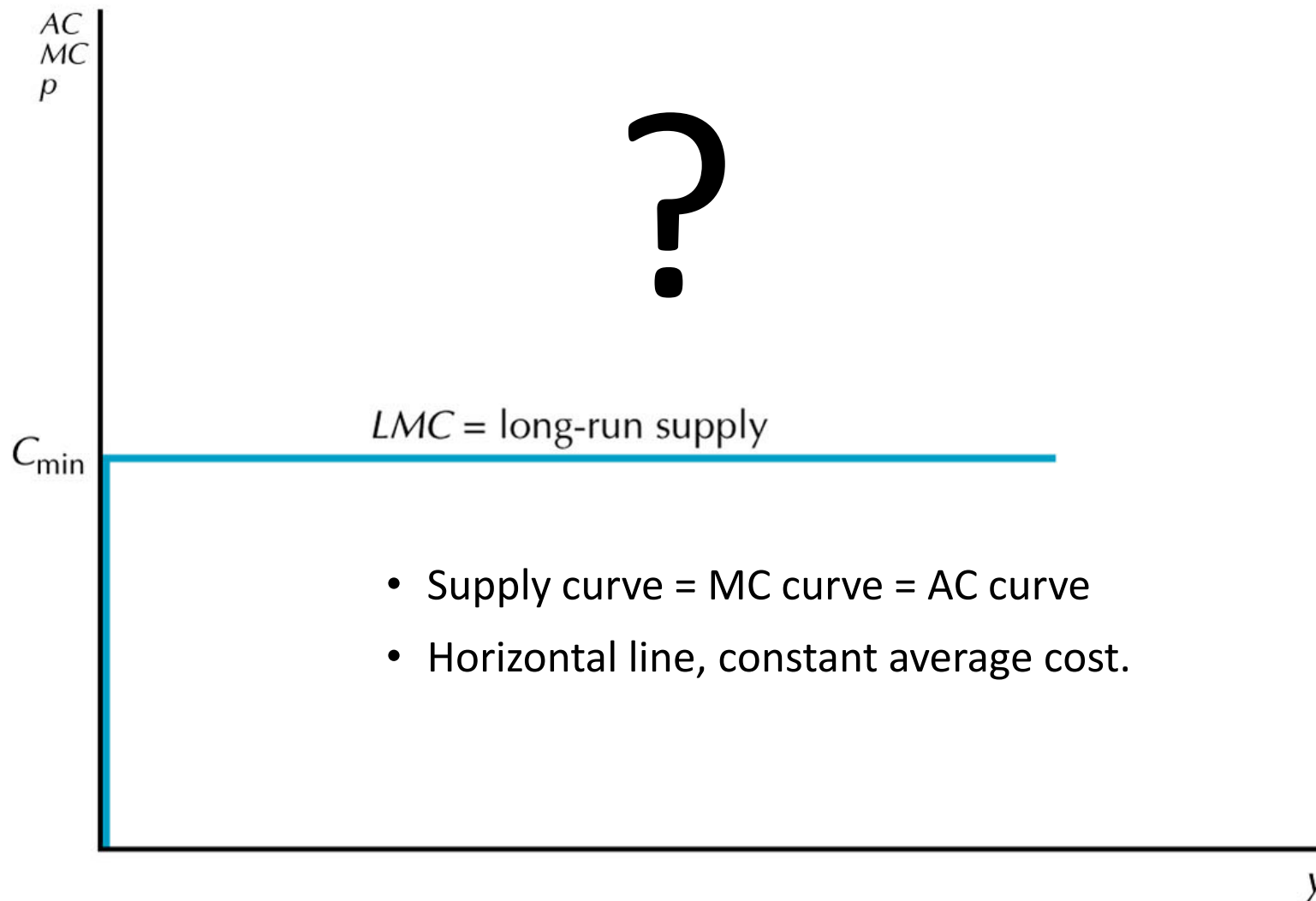
- Definition: Firm supply $S(p)$ assigns to every price the output level that maximises profit of the firm at that price
 - When firm produces strictly positive output
 - FOC requires $p = MC(y^*) \rightarrow$ inverting the marginal cost function gives firm supply
 - SOC requires that firms produce on the upward-sloping portion of the MC curve (where $\frac{dMC(y^*)}{dy} > 0$)
 - Minimum output firm is willing to produce
 - $TR(y_0) = TC(y_0)$
 - $\frac{TR(y_0)}{y_0} = \frac{TC(y_0)}{y_0}$
 - $p = AC(y_0)$
- $\rightarrow$ Firm supply curve (under perfect competition) is given by the **upward-sloping portion** of the **marginal cost curve** where **price is at least minimum average costs**

Firm supply curve



Firm supply curve: constant returns

What will the firm supply curve look like under constant returns to scale?



Profit maximisation: Exercise

Setup:

- $TC(y) = 2y^3 - 30y^2 + 150y$

Questions:

1. What output would be produced at price $p=6$?
2. Derive the inverse supply function
3. Derive the supply function



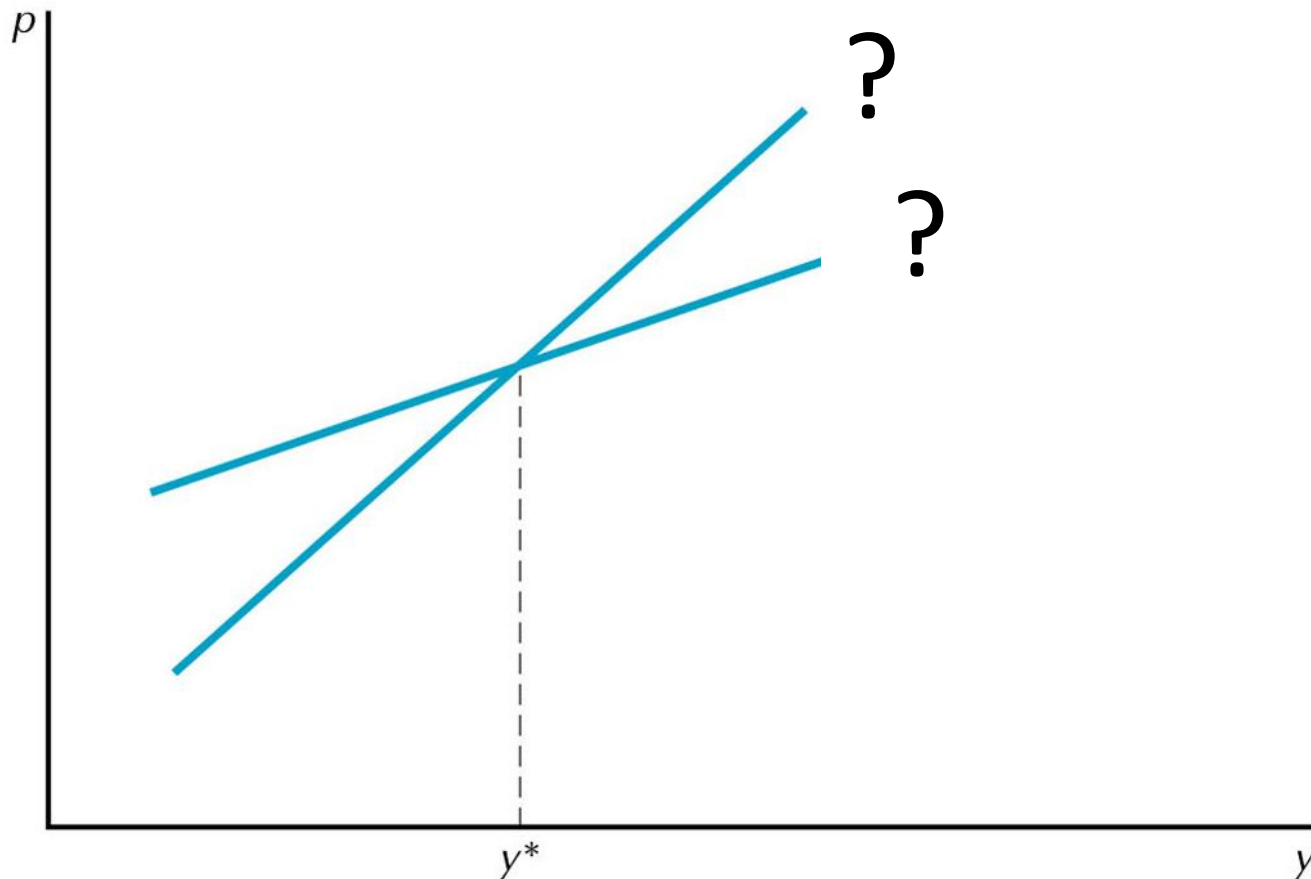
Long-run vs. short-run: adjustability of factors of production

- **Long run:** all factors of production freely adjustable – $MC(y, k(y))$
- **Short run:** some factors fixed – $MC(y, k)$
- Example: A steel mill wants to increase production in reaction to a demand increase
 - Short run: it can increase production by using more energy, more intermediate inputs and longer shifts
 - Long run: it can build new facilities and buy extra machinery



Long-run vs. short-run supply curve

→ Supply curve more elastic in the long run than in the short run



Long-run vs. short-run: variable vs. fixed costs

- There is another way to think about short run vs. long run

- **Short run**

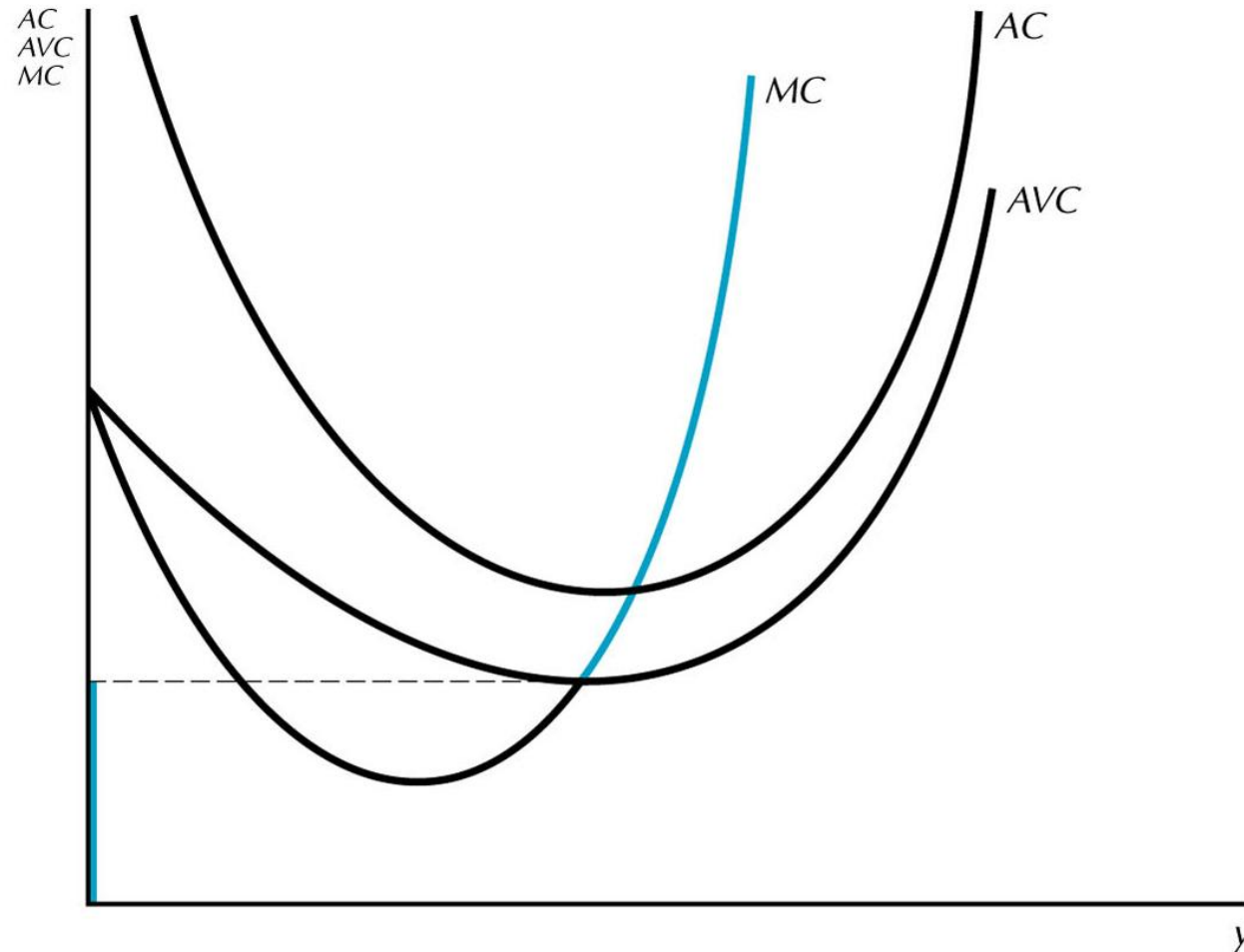
- No entry or exit
- Variable costs depend on output
- Fixed costs have to be paid
 - ➔ If $VC < py$, a firm produces and has profits $py - VC - FC$
 - ➔ If $VC > py$, a firm stops production and has profits $- FC$
 - ➔ Shut-down condition: $AVC > p$
 - ➔ Profits can be negative

- **Long run**

- A firm can exit
- Firms only operate if profits positive

Short-run supply curve

Upward-sloping portion of the marginal cost curve where price is at least minimum average **variable** costs



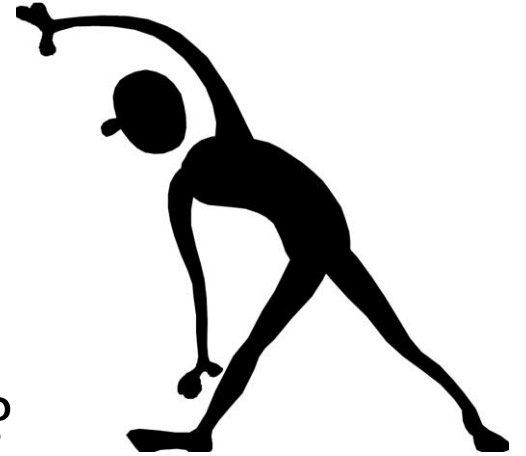
Short-run supply curve: Exercise

Setup:

- $TC(y) = y^2 + 30y + 400$
- Assume fixed costs have to be paid in the short run

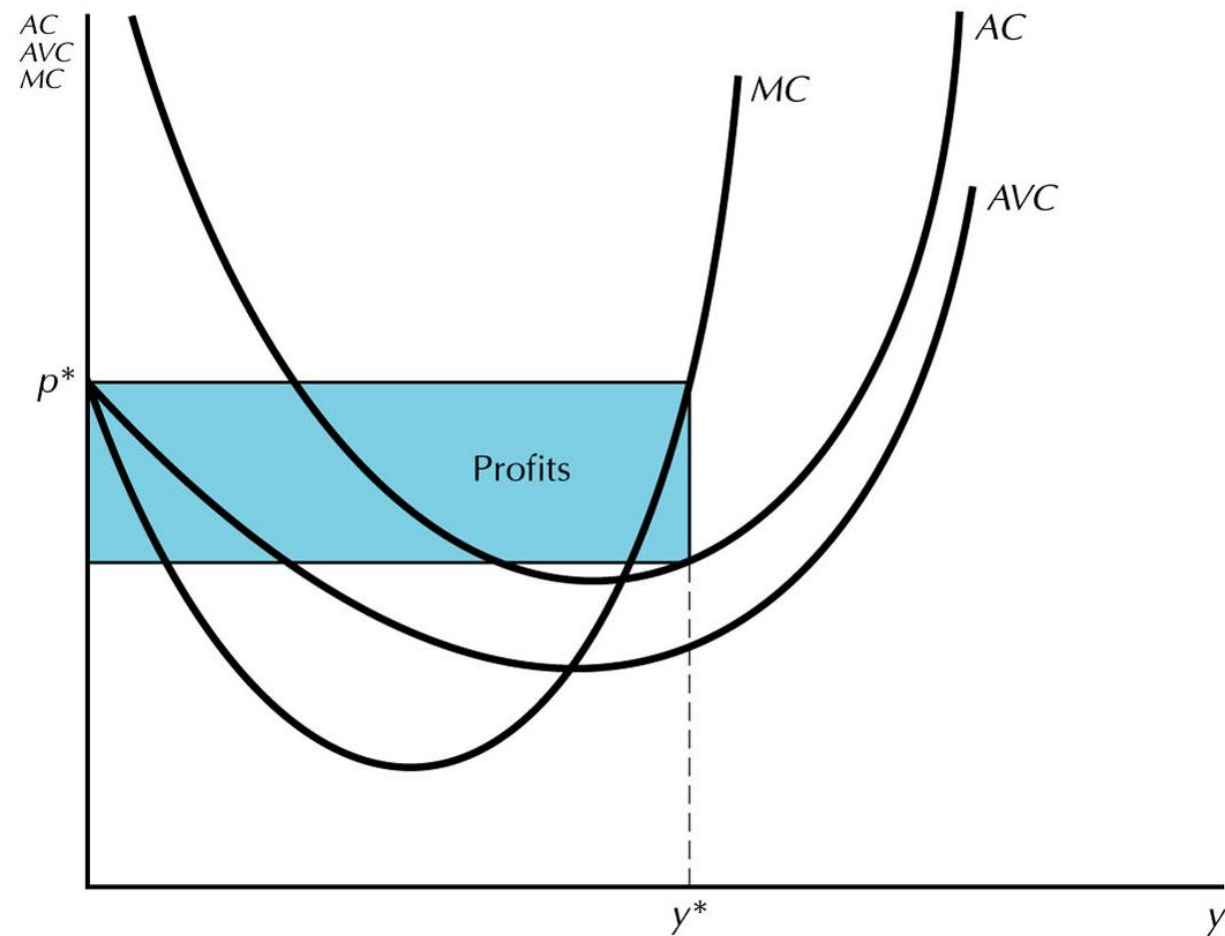
Questions:

1. What output level and price define the shut-down point?
2. Derive the short-run supply function
3. Find the point where the cost equals revenues (i.e. the firm breaks even)



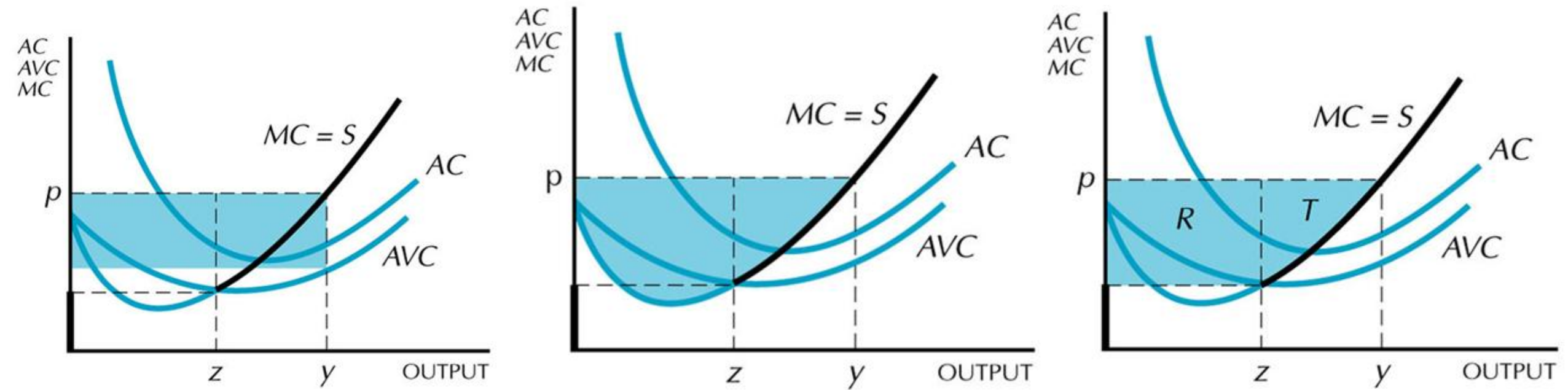
Profit

$$\text{Profit} = TR - TC$$



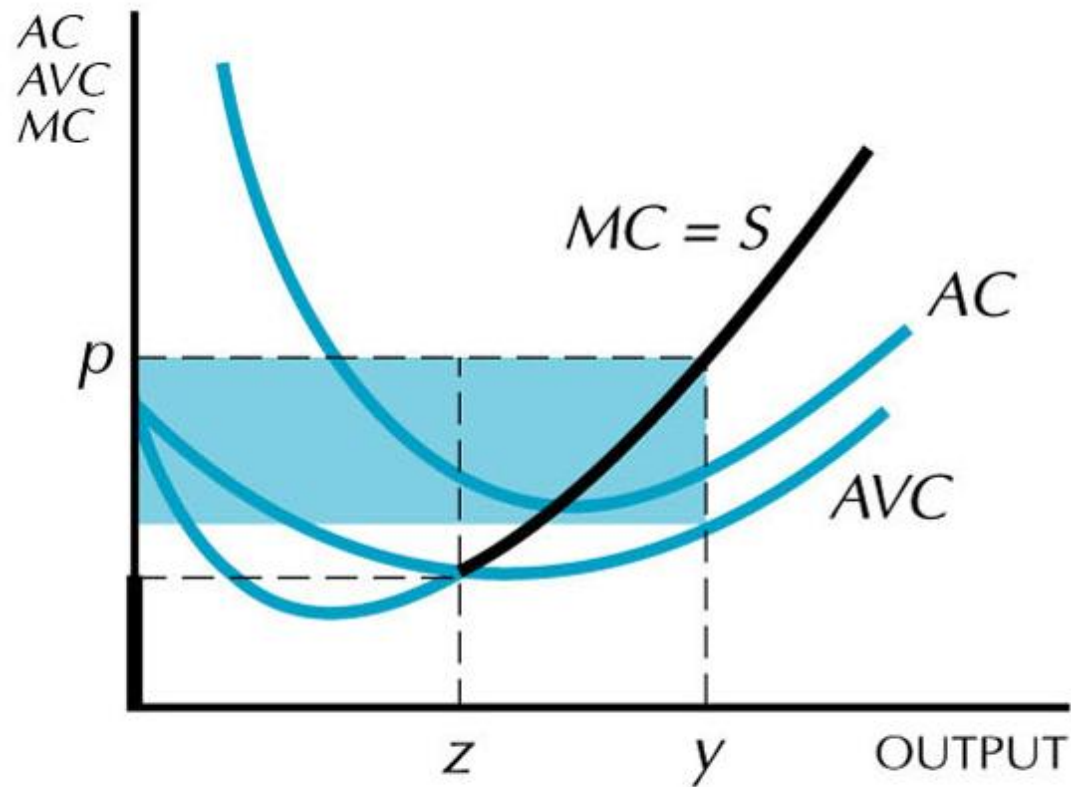
Producer's surplus

$$PS = TR - VC$$



Producer's surplus

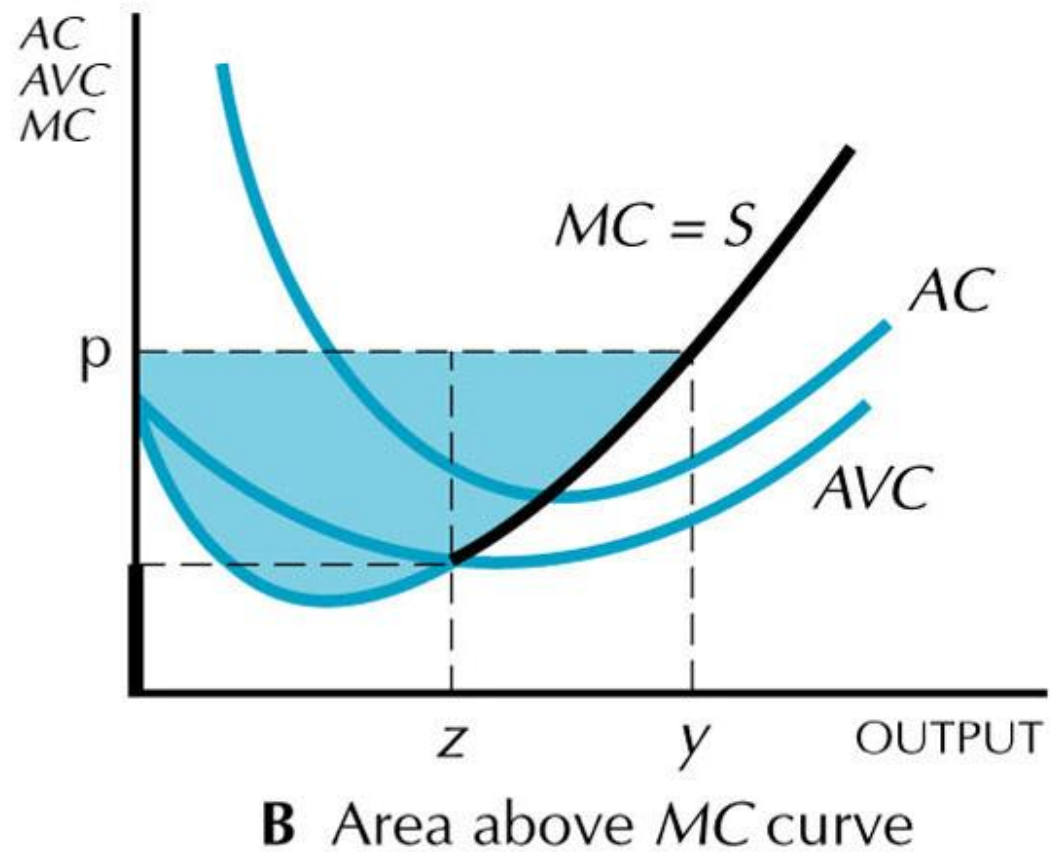
$$\mathbf{PS = TR - VC}$$



A Revenue – variable costs

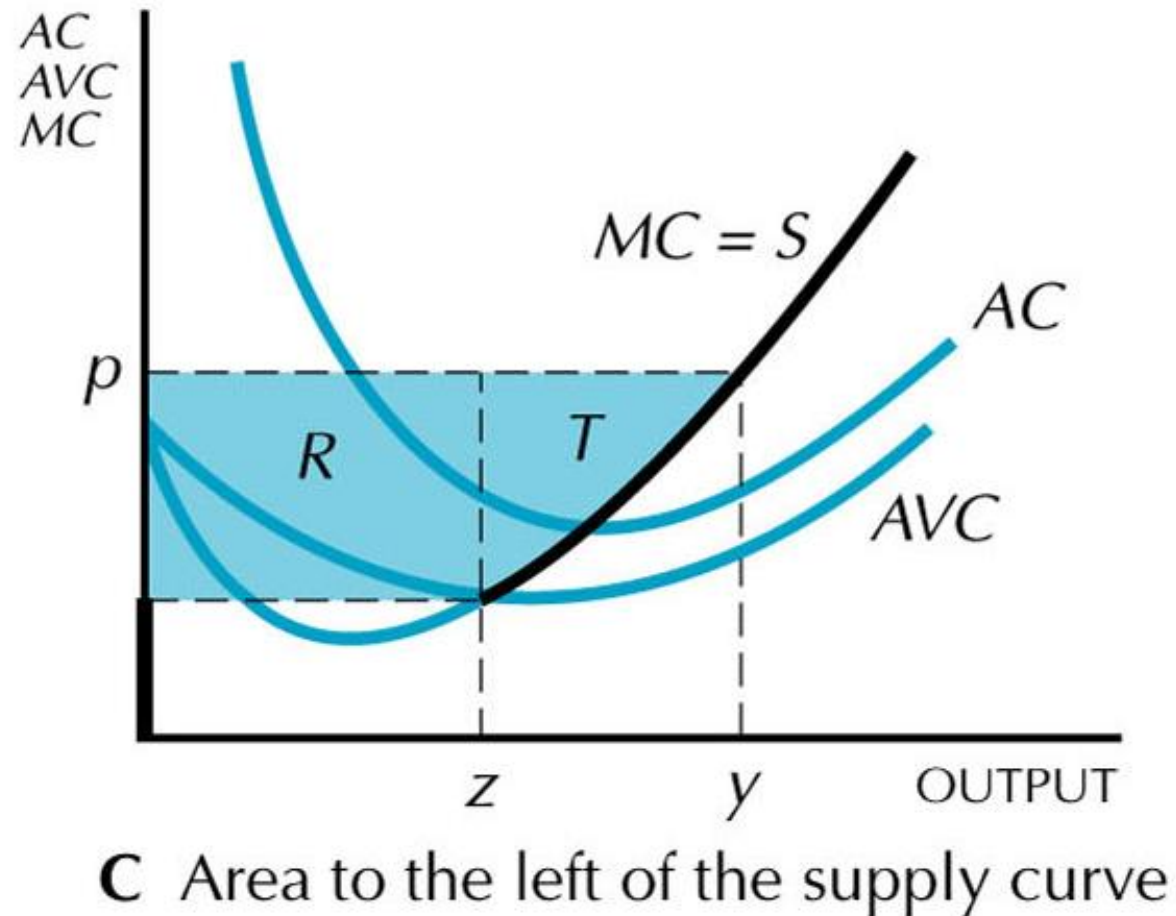
Producer's surplus

$$PS = TR - VC$$



Producer's surplus

$$\mathbf{PS = TR - VC}$$



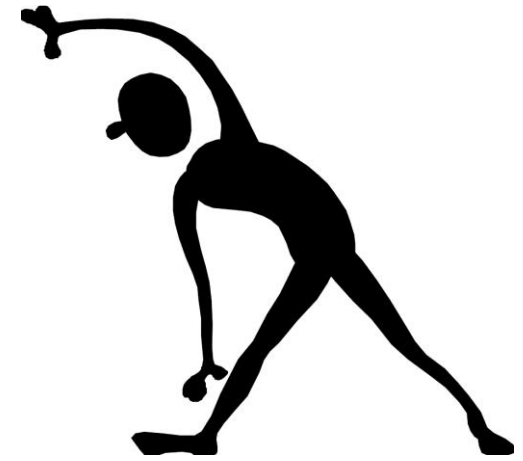
Profit & producer's surplus: Exercise

Setup

- $TC(y) = y^2 + 1$

Questions

1. Derive the supply function
2. Derive profit
3. Derive producer's surplus



Comparative statics

Marshall factor demand function $x_i^* = x_i(p, w)$

- Plug-in the supply function $y^* = y(p, w)$ into conditional demand functions $x_i^* = x_i(w, y(p, w,))$

Profit function $\pi(p, w)$

- Plug-in the supply function and Marshall factor demand functions
- $\pi(p, w) = py^* - \sum_{i=1}^N w_i x_i = py(p, w) - \sum_{i=1}^N w_i x_i(p, w) = py(p, w) - c(w, y(p, w,))$

Comparative statics: Exercise



Setup

- Production function: $f(x) = \sqrt{3x_1^{\frac{1}{3}}x_2^{\frac{2}{3}}}$
- Conditional factor demands: $x_1(w, y) = \frac{y^2}{3} \left(\frac{w_2}{2w_1}\right)^{\frac{2}{3}}$ and $x_2(w, y) = \frac{y^2}{3} \left(\frac{2w_1}{w_2}\right)^{\frac{1}{3}}$
- $c(w, y) = \frac{y^2}{2^{\frac{2}{3}}} w_1^{\frac{1}{3}} w_2^{\frac{2}{3}}$

Questions

1. Derive the supply function $y(p, w)$
2. Derive the Marshall factor demands $x_1(p, w)$ and $x_2(p, w)$
3. Derive the profit function $\pi(p, w)$

Comparative statics

Properties of the profit function

- Increasing in output price p if $y > 0$ (and non-decreasing if $y \geq 0$)
- Decreasing in input prices w_i if $x_i > 0$ (and non-increasing if $x_i \geq 0$)
- $\pi(kp, kw) = k\pi(p, w)$
- Continuous in p and w
- Smooth in p and w
- Convex in p and w
- Hotelling's lemma
 - $\frac{\partial \pi(p, w)}{\partial p} = y(p, w)$
 - $\frac{\partial \pi(p, w)}{\partial w_i} = -x_i(p, w)$

➔ Use to derive supply and Marshall demand functions from profit function

Comparative statics

Properties of the Marshall demand functions x_i

- Non-increasing in own price w_i
 - From Hotelling's lemma and convexity of the profit function:
$$\frac{\partial x_i(p, w)}{\partial w_i} = - \frac{\partial^2 \pi(p, w)}{\partial w_i^2} \leq 0$$

➔ No parallel to the Giffen good (although there are both the substitution effect and the effect of a change in output)
- $x_i(kp, kw) = x_i(p, w)$

Comparative statics

Properties of the supply function

- Non-decreasing in output price p
 - From Hotelling's lemma and convexity of the profit function:

$$\frac{\partial y(p, w)}{\partial p} = \frac{\partial^2 \pi(p, w)}{\partial p^2} \geq 0$$

- $y(kp, kw) = y(p, w)$
 - ➔ Firms do not suffer from money illusion

Group discussion

Please get in groups of 3.



1. Are crude oil markets a good example of perfect competition? Why?
2. Are firms in markets characterised by perfect competition more likely to be subject to decreasing or increasing returns to scale? Why?
3. Is the supply of electricity by ČEZ more elastic in the long run or in the short run? Why?

